Electoral Roll OCR — User Guide

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1. What This Tool Does

This tool extracts voter data from Tamil Nadu electoral roll PDFs into structured CSV files. It processes **English and Tamil PDF pairs** for each Assembly Constituency (AC), using local OCR (Optical Character Recognition) — no internet or API costs required.

What you get: A CSV file per constituency with columns for Serial No, EPIC ID, Name (English & Tamil), Relation Name (English & Tamil), Relation Type, House No, Age, and Gender.

What you need: The electoral roll PDFs downloaded from voters.eci.gov.in in both English and Tamil.

2. Prerequisites

Requirement	Details
Operating System	Windows 10/11 (primary), Linux/macOS (supported)
Python	3.10 or higher
Tesseract OCR	Version 5.4+ with Tamil language data
Disk Space	~50 KB per page pair output; ~500 MB for Tesseract + dependencies
RAM	Minimum 2 GB free (4+ GB recommended for multi-worker)

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3. Installation

Option A: Web UI Setup (Recommended)

The easiest way — the Web UI checks everything for you and installs missing components with one click.

1. Double-click `start.bat` (Windows) or run `bash start.sh` (Linux/macOS)
2. A browser window opens at `http://localhost:7000`
3. Click the **gear icon** (⚙ Setup) in the sidebar
4. The Setup panel shows three dependency checks:
   - **Python OCR Packages** — PyMuPDF, OpenCV, pytesseract, NumPy, Pillow, pypdf
   - **Tesseract Binary** — The OCR engine itself
   - **Tamil Tessdata** — Tamil language model for Tesseract
5. Click **Install** next to any item showing ❌
6. When all three show ☑, you're ready

Option B: Automated Setup (setup.bat)

```
setup.bat
```

This script checks and installs Python, Tesseract, Tamil language data, and Python packages automatically. Follow the on-screen prompts.

Option C: Manual Setup

Install Python 3.10+

Download from python.org. **Important:** Check "Add Python to PATH" during installation.

Install Tesseract OCR

```
winget install UB-Mannheim.TesseractOCR
```

During installation, check **"Additional language data"** and **"Additional script data"** to include Tamil support.

Verify:

```
"C:\Program Files\Tesseract-OCR\tesseract.exe" --version
```

Install Tamil Language Data (if not included)

Download `tam.traineddata` and copy to:

```
C:\Program Files\Tesseract-OCR\tessdata\tam.traineddata
```

Install Python Dependencies

```
pip install -r requirements.txt
```

What If Setup Fails?

Here are the most common setup failures and how to fix them:

Python OCR Packages fail to install

Problem	Solution
`pip` not found	Python wasn't added to PATH. Reinstall Python and check "Add Python to PATH"
Permission denied	Open Command Prompt as Administrator, then run `pip install -r requirements.txt`
`No module named 'fitz'`	Run `pip install pymupdf` directly
Package version conflict	Run `pip install --upgrade -r requirements.txt` to force upgrade
Build errors (C++ compiler)	Install Visual Studio Build Tools from visualstudio.microsoft.com

Tesseract fails to install

Problem	Solution
`winget` not available	Your Windows version may not have it. Download Tesseract manually from UB-Mannheim GitHub. Install the `.exe` to `C:\Program Files\Tesseract-OCR\`
Tesseract installed but not detected	Make sure `C:\Program Files\Tesseract-OCR\tesseract.exe` exists. If installed elsewhere, add its folder to your system PATH

Problem	Solution
Antivirus blocks install	Temporarily disable real-time protection, install, then re-enable

Tamil Tessdata fails to download

Problem	Solution
Permission denied on tessdata folder	The Web UI will automatically fall back to a project-local `tessdata/` folder. Alternatively, open Command Prompt as Administrator and copy the file manually
Download fails (no internet)	Download `tam.traineddata` on another machine from this link, then copy it to `C:\Program Files\Tesseract-OCR\tessdata\` via USB
Tamil not showing in `--list-langs`	The file may be corrupted. Re-download (should be ~40 MB)

Web UI itself fails to start

Problem	Solution
`fastapi` not found	Run `pip install fastapi uvicorn aiofiles psutil`
Port 7000 in use	The launcher auto-tries ports 7000-7009. If all are in use, close other applications or check with `netstat -ano | findstr :7000`
Unicode error on startup	This has been fixed in v1.4. If you see it, ensure you have the latest code
Browser doesn't open	Navigate manually to `http://localhost:7000`

General tip: If all else fails, try the Web UI Setup panel — it provides the most user-friendly installation experience with real-time progress streaming.

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4. Getting Started — Web UI

Launching the Web UI

Windows:

```
start.bat
```

Linux/macOS:

```
bash start.sh
```

The browser opens automatically at `http://localhost:7000`. This is a local-only address — nothing is exposed to the network.

Understanding the Layout

The UI has a **sidebar** on the left (desktop) or a **bottom bar** (mobile/small screens):

Section	Icon/Label	What it does
Setup	⚙ (gear)	Collapsible panel to check and install dependencies. Hidden by default — click to expand.
Workflow	▶ Workflow	Main workspace. Select an AC, configure options, run the pipeline.
Live Logs	📋 Live Logs	Watch real-time output from running jobs. Colour-coded log lines, ETA, kill button.
Data	🗀 Data	Overview of all ACs — progress, record counts, CSV download.
History	⏲ History	Browse past log files and run summaries.

5. Step-by-Step Workflow

Step 1: Prepare Input PDFs

1. Go to voters.eci.gov.in
2. Select your state (Tamil Nadu), district, and Assembly Constituency
3. Download the electoral roll PDFs in **both English and Tamil**
4. Place them in the correct folder structure:

```
Input/ER_Downloads/AC-188/
    english/    <-- English PDFs (e.g., 2026-EROLLGEN-...-ENG-1-WI.pdf)
    tamil/      <-- Tamil PDFs (e.g., 2026-EROLLGEN-...-TAM-1-WI.pdf)
```

Important: The number of English and Tamil PDFs should match. If they don't, the UI will show a yellow warning with a link to re-download.

Step 2: Create or Select an AC

If the AC already has PDFs:

- Open the **Workflow** tab
- Select your AC from the dropdown (e.g., `AC-188 (0%)`)

If you need to create a new AC:

1. Click the green `+` button next to the AC dropdown
2. Enter the AC number in `AC-xxx` format (e.g., `AC-188`)

3. The tool creates the directory structure automatically
4. Copy your PDF files into the newly created `english/` and `tamil/` folders
5. Click the ↺ refresh button to update the dropdown

Step 3: Split PDFs

What this does: Splits each multi-page PDF into individual page files. All pages are split — non-data pages (metadata, summary, maps) are automatically skipped during extraction later.

1. In the Pipeline section, find **1. Split PDFs**
2. (Optional) Check `--force` if you want to overwrite existing split files
3. Click ▶ **Run**
4. The job opens in Live Logs — wait for completion

Tip: Click the **i** button next to the step name to see a detailed explanation.

Step 4: Extract Data (OCR)

What this does: Runs Tesseract OCR on each page pair (English + Tamil) and extracts voter records into CSV files.

1. Find **2. Extract (OCR)** in the Pipeline
2. Set your desired options:
   - **Workers slider** (left panel): Controls parallel processing. The slider shows green/yellow/red based on your system resources.
   - **--part**: Leave empty for all parts, or enter a range (e.g., `1-50`)
   - **--page**: Leave empty for all pages, or enter specific pages (e.g., `1-10`)
   - **--limit**: Leave as 0 for all, or set a number to process only that many pairs
   - **--cross-check**: Enable to add validation columns to the CSV
3. Click ▶ **Run**
4. Monitor progress in Live Logs

For your first run, we recommend testing a few pages first — see Testing Before a Full Run.

Step 5: Merge Outputs

What this does: Combines individual page CSVs into part-level and AC-level files.

1. Find **3. Merge Outputs**
2. (Optional) Check `--force` if you're re-merging after additional extractions
3. Click ▶ **Run**

Step 6: Download Results

1. Go to the **Data** tab
2. Find your AC in the list
3. Click ⬇ **Download CSV** to get the merged AC-level CSV file
4. Open in Excel, Google Sheets, or any spreadsheet application

Or use Full Pipeline: Instead of running steps individually, click ⚡ **Run Full Pipeline (Split → Extract → Merge)** to run all three steps automatically.

6. Extract Options Explained

Option	Format	Example	What it does
--part	Number or range	`101`, `50-100`, `1,5,10-20`	Process only specific parts. Leave empty for all.
--page	Number or range	`4`, `1-10`, `1,5,10-20`	Process only specific pages within a part. Requires --part.
--limit	Number	`10`	Stop after processing this many pairs. 0 = no limit.
--cross-check	Checkbox	—	Cross-validates English vs Tamil data. Adds two extra columns: `Cross_Check` (OK/REVIEW) and `Cross_Check_Notes`.
--reset	Checkbox (yellow)	—	Clears the checkpoint for the specified part, allowing re-processing. **Use with care** — a confirmation dialog will appear.

Utility Buttons

Button	What it does
Dry Run	Shows what file pairs would be processed, without actually processing them. Useful to verify before committing.
Validate Page	Tests extraction on specific pages and shows detailed output in Live Logs. Respects the --part and --page settings.
Test 1 Page (bottom)	Quick one-page test with inline preview table. Great for checking OCR quality.

7. Testing Before a Full Run

Before committing to a full AC extraction (which can take hours), it's wise to test:

Quick Test (1 page with preview)

1. Select your AC
2. Click 🔍 **Test 1 Page** at the bottom of the Pipeline section
3. A preview table appears showing extracted records — verify names, EPIC IDs, ages look correct

Targeted Test (specific pages)

1. Set **--part** to a part number (e.g., `1`)
2. Set **--page** to a page range (e.g., `1-5`)
3. Click **Validate Page**
4. Check the Live Logs for detailed output

Dry Run (see what will be processed)

1. Click **Dry Run**
2. Check Live Logs — it lists all pending file pairs without processing them
3. Useful to verify the right files are queued

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8. Overnight Queue

For processing multiple ACs unattended:

1. Select an AC and configure workers/options
2. Click **Add Selected AC** in the Overnight Queue section
3. Repeat for other ACs
4. Click **Start Queue**
5. The tool runs Split → Extract → Merge for each AC sequentially
6. A browser notification fires when each AC completes

Queue features:

- Queue state is saved to disk — if the server restarts, the queue resumes
- Click **Stop After Current** to gracefully pause (finishes the current AC)
- Remove waiting items with the **X** button

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9. Monitoring Progress

During Extraction

- **Live Logs tab**: Real-time colour-coded output. Red = errors, yellow = warnings, green = success.
- **ETA indicator**: Shows percentage complete and page count (e.g., "45% done (130/290 pages)")
- **Kill button**: Red button to stop a running job immediately

Overall Progress

- **Workflow tab**: AC status card shows extraction progress bar, record count, and step completion (Downloads ✓, Split ✓, Merged ✓)
- **Data tab**: Overview of all ACs with progress bars and record counts

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10. Understanding the Output CSV

The merged CSV has **14 columns** by default:

Column	Description	Example
AC No	Assembly Constituency number	188
Part No	Part number within the AC	33
Serial No	Voter serial number	211

Column	Description	Example
EPIC ID	Voter ID card number	`RVJ1612993`
Name (English)	Voter name in English	`Kavitha`
Name (Tamil)	Voter name in Tamil	`கவிதா`
Relation Name (English)	Father/Husband name in English	`Murugesan`
Relation Name (Tamil)	Father/Husband name in Tamil	`முருகேசன்`
Relation Type	Relationship	`Father`
House No	House/door number	`1-192`
Age	Voter age	`30`
Gender	Male or Female	`Female`
DOB	Date of birth (always blank)	
ContactNo	Contact number (always blank)	

With `--cross-check` enabled, two extra columns are added:

Column	Values	Meaning
Cross_Check	`OK` or `REVIEW`	`REVIEW` = at least one field disagreed between English and Tamil
Cross_Check_Notes	Text	Details of mismatches (e.g., `EPIC mismatch EN=WXJ1234567 TA=WXJ1234568`)

File details:

- Encoding: UTF-8 with BOM (opens correctly in Excel)
- House numbers prefixed with `'` to prevent Excel auto-formatting
- Records sorted by Serial No ascending

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11. CLI Reference (Advanced)

All operations available in the Web UI can also be run from the command line:

Split PDFs

```
python split_pdfs.py --ac AC-188
python split_pdfs.py --ac AC-188 --force    # Overwrite existing
```

Extract Data

```
python extract_ocr.py AC-188 --workers 4                    # Full AC
python extract_ocr.py AC-188 --part 101                      # Specific part
python extract_ocr.py AC-188 --part 50-100                   # Part range
python extract_ocr.py AC-188 --part 3 --page 1-10            # Page range within part
python extract_ocr.py AC-188 --part 3 --page 1,5,10-20       # Specific pages
python extract_ocr.py AC-188 --limit 10 --workers 4          # First 10 pairs only
python extract_ocr.py AC-188 --validate                      # Test 1 pair
python extract_ocr.py AC-188 --part 3 --page 4 --validate    # Test specific page
python extract_ocr.py AC-188 --dry-run                        # Show pending pairs
python extract_ocr.py AC-188 --reset --part 101               # Reset checkpoint for part
python extract_ocr.py AC-188 --cross-check --workers 4        # With cross-validation
python extract_ocr.py --all --workers 4                       # All ACs
```

Merge Outputs

```
python merge_outputs.py --ac AC-188
python merge_outputs.py --ac AC-188 --force    # Re-merge all parts
```

Quality Analysis

```
python analyze_quality.py --ac AC-188
```

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12. Troubleshooting

Problem	Solution
Setup: Tesseract not found	Verify `C:\Program Files\Tesseract-OCR\tesseract.exe` exists. If installed elsewhere, add to PATH.
Setup: Tamil not available	Download `tam.traineddata` and copy to Tesseract's `tessdata` folder. See Installation.
Setup: Python packages fail	Try `pip install --upgrade -r requirements.txt`. If permission errors, run as Administrator.
Web UI won't start	Run `pip install fastapi uvicorn aiofiles psutil`. Check if port 7000 is free.
"Method Not Allowed" error	The server is running old code. Restart: close the server terminal, run `start.bat` again.

Problem	Solution
Count mismatch warning	English and Tamil PDF counts differ. Check your downloads at voters.eci.gov.in.
Tamil names are empty	Verify `tam` in `tesseract --list-langs`. Re-download `tam.traineddata` if needed.
Processing is slow	Reduce `--workers` if RAM-limited. Each worker uses ~200 MB RAM.
Merge missing pages	Use `--force` when merging: the merge does a full rewrite, not append.
Job shows error immediately	Check Live Logs — the subprocess exit message shows the cause.
Server changes not taking effect	Restart the web server. Python code changes require a server restart.
Grid detection failed	Automatic fallback handles this. If data quality is poor, try `--validate` on the specific page.

13. FAQ

Q: Does this tool need internet access? A: No. Once installed, everything runs locally. The only time internet is needed is for initial setup (downloading Tesseract and Python packages).

Q: How long does processing take? A: About 60 seconds per page pair. A full AC with ~720 pairs takes 3-4 hours with 4 workers. Multiple ACs can be queued for overnight processing.

Q: Is my data sent anywhere? A: No. All processing happens on your machine. No data leaves your computer.

Q: Can I stop and resume later? A: Yes. The tool saves a checkpoint after each page pair. When you run extraction again, it picks up where it left off automatically.

Q: What if I need to re-process some pages? A: Use `--reset` with `--part` to clear the checkpoint for specific parts. Or use `--page` with `--validate` to re-inspect individual pages without affecting the checkpoint.

Q: What accuracy can I expect? A: 99.87% cell-level accuracy based on validation of 1,118 records. Occasional EPIC ID letter/digit confusion may occur due to source PDF resolution (115 DPI).

Q: Can multiple people work on different ACs simultaneously? A: Yes. Each AC has its own checkpoint. Different users can process different ACs at the same time (on different machines or terminals).

Q: What's the `--cross-check` option for? A: It re-reads Tamil cells to independently verify EPIC IDs, house numbers, and serial numbers against the English data. Records with disagreements are flagged as `REVIEW` — useful for quality auditing.

Q: How do I update to a newer version? A: Pull the latest code (if using git) or download the updated files. Restart the web server. Your existing data and checkpoints are preserved.

This guide covers the Electoral Roll OCR Tool v1.4. For technical details, see [README.md](). For developer reference, see [CLAUDE.md]().